

Problem Set 1

Due on Gradescope: check the course Drive calendar

Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

- [1. \[14 points total\] Short questions with capital answers.](#)
- [2. \[12 points total\] Cobb-Douglas Come Out to Play?](#)
- [3. \[16 points total\] So how low can we go, Solow? A capital response to immigration?](#)

Did you work with other students? List them below:

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

“As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.”

Type your name:

1. [14 points total] Short answers with capital answers.

- (a) [2 points] Recall that GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \quad (1.1)$$

Roughly how big are each of these components as a *percentage* of GDP in the U.S. today?

- (b) [2 points] What exactly does investment spending create? Explain.

- (c) [2 points] In the first half of ECON 100B, what is capital? Describe it in a few sentences. Maybe describe the economic meanings of the word that do NOT directly apply to our study of economic growth.

- (d) [2 points] Recall that real GDP Y^R equals nominal GDP Y^N divided by the price level:

$$Y^R = \frac{Y^N}{P} \quad (1.2)$$

Take the growth rate of both sides of equation (1.2), and recall that the growth rate of the price level is inflation and denoted by a lowercase Greek letter “pi,” π . If the growth rate in nominal GDP is 5% and inflation is 2%, what is the growth rate of real GDP?

- (e) [2 points] In Chapter 4, we will see that GDP is modeled well by this Cobb-Douglas production function:

$$Y = A K^\alpha L^{1-\alpha} \quad (1.3)$$

where α is a constant number between 0 and 1 that is also capital’s share of income. In the textbook, $\alpha = 1/3$, which is an appropriate level of the parameter for U.S. data.

Take the growth rate of both sides of equation (1.3), treating A , K , and L as variables that potentially have growth rates. Simplify and briefly describe your answer:

- (f) [2 points] Using your answer above, describe what happens to Y if A is constant while K and L both increase by 2%. Be specific and briefly explain.

- (g) [2 points] Using your answer above, describe what happens to Y if A grows by 3% while K and L both increase by 2%. Be specific and briefly explain.

2. [12 points total] Cobb-Douglas Come Out to Play?

Although there are some phenomena in economics that the Cobb-Douglas function cannot capture realistically, it is a remarkably versatile tool that you will likely see again and again in your undergraduate studies. Recall that in Chapter 4, we see that the U.S. macroeconomy can be modeled well using Cobb-Douglas production:

$$Y = A K^{\alpha} L^{1-\alpha} \quad (2.1)$$

where $\alpha = 1/3$ is capital's share of income. When we take first partial derivatives of equation (2.1) we reveal the useful functional forms of the marginal products of capital (MPK) and of labor (MPL), which in competitive markets equal their factor prices: the interest rate r and the wage w :

$$\frac{\partial Y}{\partial K} = MPK = r \quad r = \frac{1}{3} A \left(\frac{L}{K} \right)^{2/3} \quad (2.2)$$

$$\frac{\partial Y}{\partial L} = MPL = w \quad w = \frac{2}{3} A \left(\frac{K}{L} \right)^{1/3} \quad (2.3)$$

- (a) [2 points] Take the growth rate of both sides of equation (2.2), which shows r as a function of A , K , and L , and simplify. If A is constant, then what must be true about the growth rate of r when labor and capital are growing at the same rate and thus the capital-labor ratio is constant? Show your work.

- (b) [2 points] Using your answer above, what happens to r if there is no growth in A but K rises above L by 1%?

Consider what happens when we take the natural log of both sides of equation (2.3), which is similar to taking the growth rate:

$$\log w = \log \frac{2}{3} + \log A + \frac{1}{3} \log K - \frac{1}{3} \log L \quad (2.4)$$

- (c) [3 points] In the **labor market**, assume technology A and capital K are both fixed. On the axes below, graph the simple relationship that you see in equation (2.4) between the log wage on the vertical axis and $\log L$ on the horizontal axis. Explain what you drew. Report the numerical values of the vertical intercept and the slope.



- (d) [2 points] In the picture you just drew, add a vertical **labor supply curve** and show the equilibrium wage and employment in the labor market. Briefly explain below.

- (e) [3 points] Redraw the labor market on the axes below and depict and describe what would happen in the labor market if the amount of **capital** in the economy were to increase, other things equal.

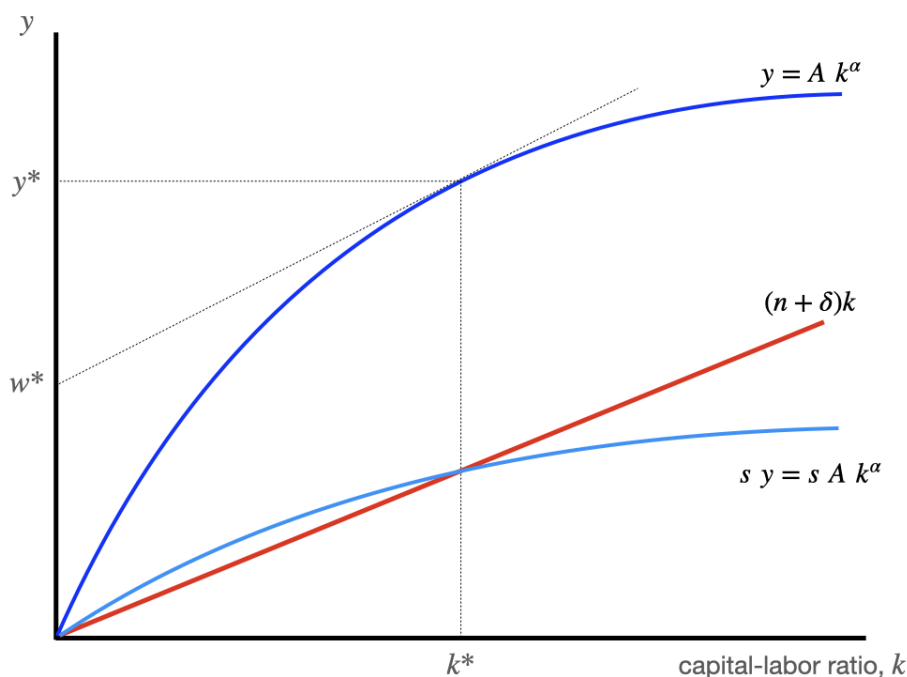


3. [16 points total] So how low can we go, Solow? A capital response to immigration?

In the standard neoclassical growth model of Robert Solow without technological growth, consumers supply labor inelastically and save out of income at rate s . There are constant returns to scale in labor and capital, and productivity A is fixed and not changing. Capital depreciates at rate δ , and labor increases at the rate of population growth, n . When output is Cobb-Douglas,

capital per worker is	$k = K/L$	also called the “capital-labor ratio”
output per worker is	$y = A k^\alpha$	and typically $\alpha = 1/3$
saving per worker is	$s y = s A k^\alpha$	
depreciation per worker is	$(n + \delta)k$	

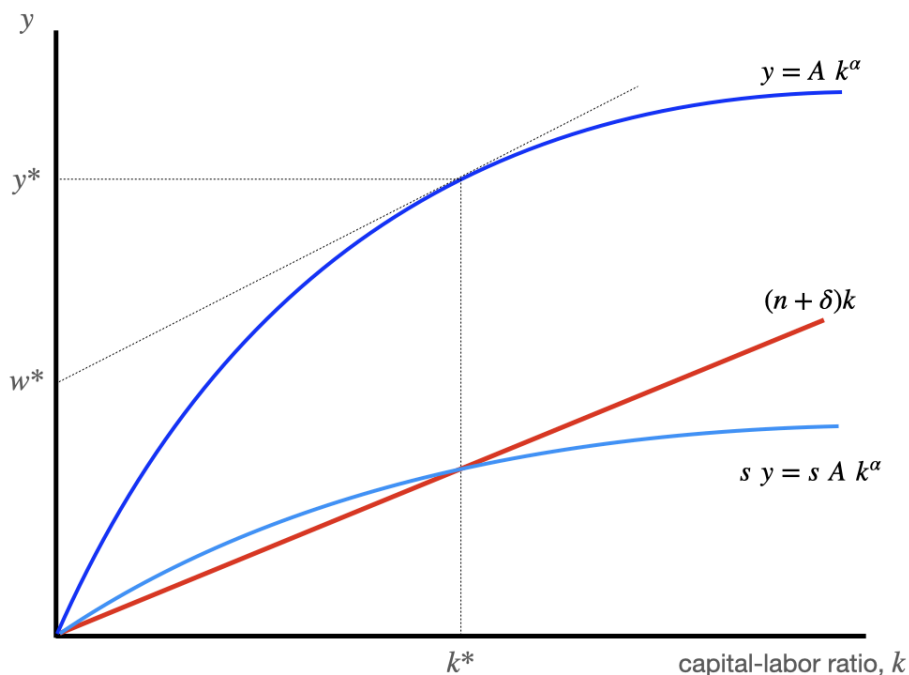
The Solow diagram below shows how capital per worker reaches a *steady state* at k^* , where depreciation per worker equals saving per worker: $(n + \delta)k = s y$, so capital per worker is unchanging there. The steady state is shown by the vertical dashed line in the diagram below.



- (a) [2 points] If the capital per worker is in steady state at k^* and not changing over time, what is true about the wage w^* , which is shown in the graph on the y -axis?

Suppose that the economy is in steady state, and then the government carries out a large removal of immigrant workers, one that increases capital per worker k by about 25%. The policy removes workers but leaves capital K and productivity, A , unchanged.

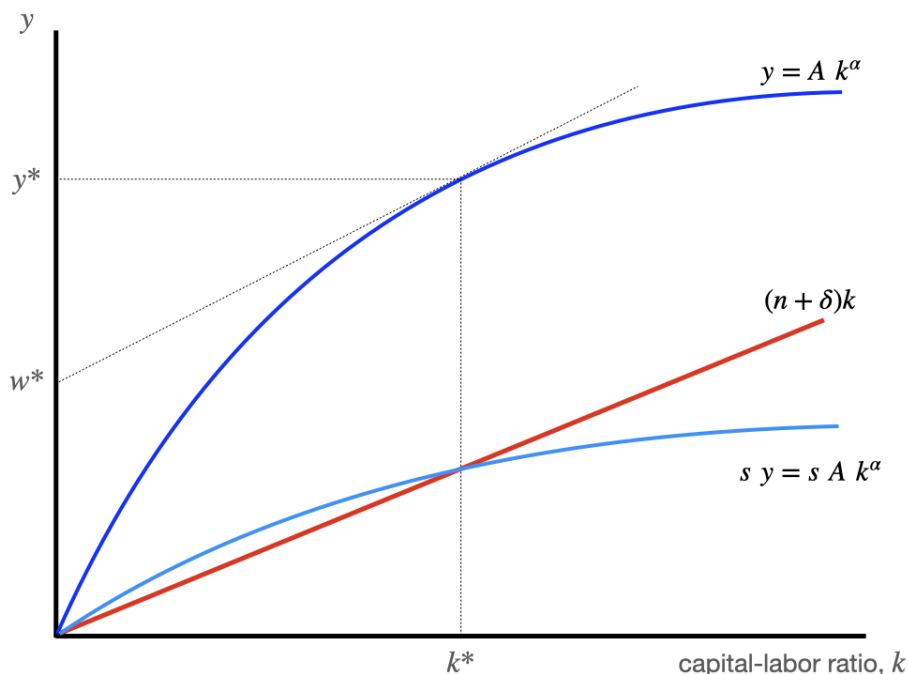
- (b) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n , δ , s , A , and α change?]



- (c) [2 points] A higher level of capital per worker will produce a higher wage, because workers are more productive. And removing any type of worker will raise capital per worker. But is that change temporary or permanent? Why?

Now suppose that while in steady state, the government instead decides to change **immigration policy**, reducing number of refugee and work visas each year, which reduces the rate of population growth, n , to a new level $n' < n$. (Assume there is no removal policy.)

- (d) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n , δ , s , A , and α change?]



- (e) [4 points] What would have happened in the graph in part (d) if immigration enhanced productivity, A , so that reducing immigration reduced the level of productivity? Explain.